



# Test Plan and Test Case

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# 1. Test Plan

- A **TEST PLAN** is a document describing software testing scope and activities. It is the basis for formally testing any software / product in a project.
  - the scope, approach, resources and schedule of intended test activities.
- **Master Test Plan:** A single high-level test plan for a project / product that unifies all other test plans.
- **Testing Level Specific:** Test plans for each level of testing:
  - Unit Test Plan
  - Integration Test Plan
  - System Test Plan
  - Acceptance Test Plan
- **Testing Type Specific:** Test plans for specific types of testing like Performance Test Plan and Security Test Plan.

## Test Planning Process

- 1) Define test strategy
- 2) Define test system
- 3) Estimate test effort
- 4) Prepare and review test plan

# 1) Test Strategy

- **Scope**
  - You cannot test everything.
  - What are the priorities?
    - most important features
    - new code
    - most difficult areas
    - most often used features
- **Approach:** how to do different tasks
- **Entry/exit criteria**
  - When can you start?
  - When can you stop?
  - When can you restart?
- **Automation**
  - Which tools?
  - When? (e.g., only regression testing)

## 2) Test Systems

- **Architecture**
  - **Test suite**: collection of tests for a related group of requirements
  - **Test case**: smallest unit
- **Environment**: physical facilities , platforms

### 3) Estimating Test Effort

- **Tasks:** list of things to do
- **Effort estimation:** previous experience, expert judgment, decomposition
- **Schedule:** show relationships between tasks
- **Risks:** estimate likelihood and impact

# Test Plan Components (1)

The format and content of a software test plan vary depending on the **processes**, **standards**, and **test management tools** being implemented.

The following format, which is based on IEEE standard for software test documentation, provides a summary of what a test plan can / should contain.

1. ***Test plan identifier***: unique
2. ***Introduction***: objectives of document/Specify constraints
3. ***Test items***: high-level description of inputs to testing  
(software / products and their versions)
4. ***Features to be tested***: list of features or requirements
5. ***Features not to be tested***: features you cannot test yet



## Test Plan Components (2)

### 6. *Approach*

- how you will test
- may include: phases of testing (unit, system, acceptance), static vs. dynamic methods, performance testing, regression testing, use of problem tracking system

### 7. *Item pass/fail criteria*

- When does a test pass?
- When does the product pass?

## Test Plan Components (3)

- 8. ***Suspension criteria and resumption requirements***: what to do if bugs prevent progress
- 9. ***Test deliverables***: all outputs of testing  
Test Case/Defects Report/ Test Report...
- 9. ***Environmental needs***: hardware and software/network  
any testing or related tools.
- 10. ***Responsibilities***: who does what/each team / role / individual.

## Test Plan Components (4)

- 13. *Staffing and training needs*:** list people or types of people
- 14. *Schedule*:** milestones
- 15. *Risks and contingencies*:** matrix here
  - the risks that have been identified.
  - the mitigation plan and the contingency plan for each risk.
- 16. *Approvals*:** signatures

## 2. Test Case

- A **TEST CASE** is a documented set of preconditions (prerequisites), procedures (inputs / actions) and postconditions (expected results) which a tester uses to determine whether a system under test satisfies requirements or works correctly.
- A test case can have one or multiple test scripts .
  - A sequence of instructions for the execution of a test
- A collection of test cases is called a test suite.

